

SUBSYSTEM REQUIREMENTS SPECIFICATION

Digital Field Elements
DCA - IO

Hitachi Rail – Internal

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LOG OF CHANGES			
Edition	Date	Author	Change Description
01PD01	2025-11-20	Hitachi Rail GTS Austria / EJH ARC E. Brenner Hitachi Rail GTS Austria / EJT RM G. Eber	First edition. Details see 7.1.
01PD02	2025-12-12	Hitachi Rail GTS Austria / EJH ARC E. Brenner Hitachi Rail GTS Austria / EJT RM G. Eber	Second edition. Details see 7.2.

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1. INTRODUCTION

1.1 PURPOSE

The present document contains the subsystem specification requirements applicable to the DCA module of the Digital Interlocking (DIL) project for the customer ÖBB (Austrian Railroad Company). It is developed in the context of the “Generic Application Digital Interlocking OBB” (DIL).

This subsystem requirement specification contains the requirements for the DCA IO.

1.2 INTENDED AUDIENCE

- DCA Development Team
- Product Design Authority
- Technical Project Managers
- Solution Engineering Managers
- Product Line Manager
- Test Manager
- Validator
- Verifier
- Safety Assurance Manager
- Project Security Manager
- Quality Assurance Manager
- RAM Assurance Manager
- System Architects

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2

1.4 DOCUMENT MANAGEMENT

See Requirements Management Plan [DIL_RMP] for a description of the SSS document management.

1.5 REFERENCED DOCUMENTS

[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed. 01
[CRS_RAMs]	D2.2 CRS RAMS - Version 1.3 - 21.05.2025
[CRS_GenReq]	D2.3 CRS General Requirements - Version 1.3 - 21.05.2025
[CRS_Sec]	D2.4 CRS Security - Version 1.4 - 07.11.2025
[CRS_P]	D2.6 CRS Subsystem Point - Version 1.3 - 21.05.2025
[CRS_LS]	D2.7 CRS Subsystem Light Signal - Version 1.4 - 07.11.2025

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[CRS_IO]	D2.9 CRS Subsystem Generic I/O - Version 1.2 - 21.05.2025
[CRS_TDS]	D2.10 CRS Subsystem TDS - Version 1.4 - 07.11.2025
[CRS_FES]	D2.11 CRS Field Element Station - Version 1.1 - 03.07.2024
[CRS_IT_PiO]	2025-11-27_OBB_RA-DSTW-D9_IT-Inbetriebnahme_Konzept_V1.3 en-GB
[CRS_SEC_CONCEPT]	2025-11-27_OBB_RA-DSTW-D8_Security_Konzept_V1.2
[OC_Elec_Draw]	OC_Electrical_Drawings - EL-SYS-25-530, Ed.01
[CRS_DDSP_Con]	2025-05-12_OBB_RA-DSTW-D11_DDSP-Konzept_V1.7
[CRS_DDSP_UC]	OBB_RA-DSTW-D11_DDSP-Systemfunktionen_V1.7
[SSS_DIL_GEN]	System Segment Specification DIL - General Requirements 3BU 17300 1251 DTAPA, Ed.01
[SSS_DIL_CU]	System Requirements Specification - Digital Interlocking OBB – DIL-CU Requirements, 3BU 17300 1255 DTAPA, Ed.01
[SSS_DIL_FES]	System Requirements Specification - Digital Interlocking OBB – Field Element Controller and Stations, 3BU 17300 1256 DTAPA, Ed.01
[SSS_DIL_RAMs]	System Requirements Specification - Digital Interlocking OBB – RAMS Requirements, 3BU 17300 1257 DTAPA, Ed.01
[SSS_DIL_SEC]	System Segment Specification DIL - Cybersecurity 3BU 17300 1252 DTAPA, Ed.01
[EULYNX Specification]	EULYNX Baseline Set 4 Release 4
[RAM_PLAN]	Project specific RAM Plan - Digital Field Elements (DFE) 3BU_62800_0018_DUAPA, Ed.02
[SAFETY_PLAN]	Safety Plan DSTW, 3BU_17300_1204_DUAPA, Ed.01

Table 1: Reference Documents

1.6 STANDARDS AND PROCEDURES

[EN 50121-1]	EN 50121-1:2017 Railway applications. Electromagnetic compatibility General: 2017-12
[EN 50121-4]	EN 50121-4:2016 + A1:2019 (OVE EN 50121-4:2017 + A1:2020) Electromagnetic compatibility – Emission and immunity of the signalling and telecommunications apparatus
[EN 50122]	EN 50122-1:2022 Railway applications - Fixed installations - Electrical safety, earthing and the return circuit
[EN 50124-1]	EN 50124-1:2017 (OVE EN 50124-1:2018 + AC:2023) Railway applications - Insulation coordination - Part 1: Basic requirements - Clearances and creepage distances for all electrical and electronic equipment
[EN 50125-3]	EN 50125-3:2003 + AC:2010 (ÖVE/ÖNORM 50125-3:2003 + AC:2010) Railway applications - Environmental conditions for equipment - Part 3: Equipment for signalling and telecommunications
[EN 50126]	EN 50126-1:2017 + A1:2024 (OVE EN 50126-1:2025) Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process EN 50126-2:2017 + A1:2024 (OVE EN 50126-2:2018 + A1:2025) Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety
[EN 50129]	EN 50129:2018 + AC:2019 (OVE EN 50129:2019) Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling

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[EN 50159]	EN 50159:2010 + A1:2020 (ÖVE/ÖNORM EN 50159:2011 + A1:2020) Railway applications - Communication, signalling and processing: Safety-related communication in transmission systems
[EN 50716]	EN 50716:2023 (OVE EN 50716:2024 + AC:2025) Railway Applications - Requirements for software development
[EN 60529]	EN 60529:1991 + AC:1993 + A1:2000 + A2:2013 + AC:2016 + AC:2019 (ÖVE/ÖNORM EN 60529:2014-11 + AC:2017 + AC2:2019) Degrees of protection provided by enclosures (IP Code)
[EN 60664-1]	IEC 60664-1:2020 + AC:2020 (OVE EN IEC 60664-1:2022) Insulation coordination for equipment within low-voltage supply systems -Part 1: Principles, requirements and tests
[EN 60721-3-1]	EN IEC 60721-3-1:2018 (OVE EN IEC 60721-3-1:2019) Classification of environmental conditions Part 3-1: Classification of groups of environmental parameters and their severities - Storage
[EN 60721-3-2]	EN IEC 60721-3-2:2018 + AC:2018 + AC:2022 (OVE EN 60721-3-2:2019 + AC:2023) Classification of environmental conditions Part 3-2: Classification of groups of environmental parameters and their severities Transportation (IEC 104/710/CDV)
[EN 60721-3-3]	EN IEC 60721-3-3:2019 (OVE EN IEC 60721-3-3:2020) Classification of environmental conditions – Part 3-3: Classification of groups of environmental parameters and their severities – Stationary use at weather protected locations
[EN 61010-1]	EN 61010-1:2010 + A1:2019 + AC:2019 (OVE EN 61010-1:2020 + AC:2022 + AC2:2023) Sicherheitsbestimmungen für elektrische Mess-, Steuer-, Regel- und Laborgeräte - Teil 1: Allgemeine Anforderungen
[EN 62368-1]	EN IEC 62368-1:2024 + A11:2024 (OVE EN IEC 62368-1:2025) Einrichtungen für Audio/Video-, Informations- und Kommunikationstechnik -Teil 1: Sicherheitsanforderungen / Audio/video, information and communication technology equipment -Part1: Safety requirements
[EN 60332-1-2]	EN 60332-1-2:2004 + A11:2016 + A12:2020 + A1:2015 (OVE EN 60332-1-2:2022) Tests on electric and optical fibre cables under fire conditions - Part 1-2: Test for vertical flame propagation for a single insulated wire or cable – Procedure for 1 kW pre-mixed flame
[EN 60332-2-2]	EN 60332-2-2:2004 (ÖVE/ÖNORM EN 60332-2-2:2005) Tests on electric and optical fibre cables under fire conditions - Part 2-2: Test for vertical flame propagation for a single small insulated wire or cable - Procedure for diffusion flame
[EN 60754-1]	EN 60754-1:2014 + A1:2020 (OVE EN 60754-1:2021) Test on gases evolved during combustion of materials from cables – Part 1: Determination of the halogen acid gas content
[EN ISO 9223]	EN ISO 9223:2012 (ÖNORM EN ISO 9223:2012) Korrosion von Metallen und Legierungen - Korrosivität von Atmosphären - Klassifizierung, Bestimmung und Abschätzung
[EN 60068-2-64]	EN 60068-2-64:2008 + A1:2019 (OVE EN 60068-2-64:2020) Environmental testing - Part 2-64: Tests - Test Fh: Vibration, broadband random and guidance
[EN IEC 62443-4-1]	EN IEC 62443-4-1:2018 (OVE EN IEC 62443-4-1:2018) Security for industrial automation and control systems, Part 4-1: Secure product development lifecycle requirements
[EN IEC 62443-4-2]	EN IEC 62443-4-2:2019 + AC:2022 (OVE EN IEC 62443-4-2:2020) Security for industrial automation and control systems – Part 4-2: Technical security requirements for IACS components

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[OEBB R10]	ÖBB Merkheft R10, Stand 01.01.2022 Eisenbahnanlagen: Schwerpunktkonzept aus Sicht des Arbeitnehmer/innenschutzes
[RoHS]	Directive 2011/65/EU of the European Parliament and of the Council of 8 June 2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (recast). Consolidated version of 24 June 2015.
[REACH]	Regulation (EC) No 1907/2006 - REACH of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) and establishing a European Chemicals Agency

Table 2: Standards and Procedures

1.7 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

ADM	Administration Diagnostic Monitoring
BDI	Basic Data Identifier
CAN	Controller Area Network
CENELEC	European Committee for Electrotechnical Standardization
CPU	Central Processing Unit
DCA	Distributed Controller Architecture
DDSP	Data Diagnosis Service Platform (MDM+SSP by ÖBB)
DFE	Digital Field Elements
DIL	Digital Interlocking
DIL-CU	Digital Interlocking Central Unit
DNS	Domain Name System
ECMUX	Element Controller Multiplexer
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMS	Electromagnetic Susceptibility
FES	Field Element Station
FRACAS	Failure Reporting, Analysis and Corrective Action System
Gen-IO	General Input Output
HW	Hardware
IO	Input Output
IP	Ingress Protection / Protection class
IP	Internet Protocol
IPv4	Internet Protocol, Version 4
IT	Information Technology
LED	Light-Emitting Diode
LH	Lastenheft (Customer Requirements)
LS	Light Signal
MAC	Media Access Control (MAC address)
MRT	Mean Repair Time
MTTF	Mean Time To Failure
MTTR	Mean Time To Recover
NTP	Network Time Protocol
OC	Object Controller
OSI	Open Systems Interconnection (OSI model)
P	Point
PKI	Public Key Infrastruktur
PoS	Point of Signalling
QoS	Quality of Service

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QR	Quick Response (QR Code)
RAMS	Reliability, Availability, Maintainability, Safety
RMP	Requirements Management Plan
SCI-IO	Standard Communication Interface Input / Output
SDI-IO	Standard Diagnostic Interface Input / Output
SiDiNet	Sicherungsdienstnetzwerk - Signalling Network
SIL	Safety Integrity Level
SMI-IO	Standard Maintenance Interface Input / Output
SSI-IO	Standard Security Interface Input / Output
SW	Software
TDS	Train Detection System
THR	Tolerable Hazard Rate
TLS	Transport Layer Security
UV	Ultra Violet
VIL	Virtual Inter Locking
VLAN	Virtual Local Area Network

Table 3: Abbreviations and Acronyms

1.8 STRUCTURE OF THE REQUIREMENTS

See [DIL_RMP] for a description of the SSS requirement structure, the used attributes and their meaning.

1.9 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS.

Settings applied:

DOORS/RMF Version:	DOORS 9.7.2.6/RMF 6.1.0.9
DOORS/RMF Module Name:	Helper für EL4 SSS
DOORS/RMF Module Version (Baseline):	0.0
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS 0013 (DCA-IO)
DOORS/RMF Current Filter:	(((((Include Object 201) OR (Absolute Number == 199)) OR (Absolute Number == 200)) OR (Absolute Number == 201)) OR (Absolute Number == 226)) OR (Absolute Number == 227))
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostprocessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1258.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1258.log,true,true,false,false,false,false,0,true,4,
DOORS/RMF Export Options 4:	3,###,==,true,false,false,false,false,
Documentation-Vob Template Name:	3BU_62800_0013_DTAPA-Template

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1.10 OPEN POINTS

Chapter 1.6 will later be linked to a separate glossary document for the entire DSTW.
Review values concerning “Allocation” with architect/design team.
Review values concerning “Verification” with test manager.
Check completeness of attribute “Origin”.
The security requirements regarding Basic Data Identifier will be specified in more detail during Security Threat Assessment.

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the VIL/ECMUX. The Object Controllers is installed in Field Element Stations (FES) supplied by Hitachi Rail Austria.

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3. COMMON REQUIREMENTS FROM SYSTEM SPECIFICATION [SSS_DIL_GEN]

3.1 GENERAL REQUIREMENTS

3.1.1 ENVIRONMENT AND PROTECTION

3.1.1.1 ELECTROMAGNETIC COMPATIBILITY (EMC)

EL4-SSS-REQ-00601		EMI - EN 50121-4		(Accepted)
The [DIL] system, its subsystems, as well as components close to the track shall be compliant with the limits for EMI according to [EN 50121-4].				
Comments:				
Category:	PER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_248]	
Target Release:	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:		
Verification:	Qualification test			
Issues/Decisions:				

EL4-SSS-REQ-00603		EMS - EN 50121-4		(Accepted)
The EMS testing of the subsystems close to the track, shall be performed according to the conditions listed in [EN 50121-4] chapter 6.2 (interference immunity, whereby assessment criterion A in accordance with [EN 50121-1] shall be used as the basis for all limit values).				
Comments:				
Category:	TVR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_250]	
Target Release:	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:		
Verification:	Qualification test			
Issues/Decisions:				

3.1.1.2 VIBRATIONS AND SHOCKS

A suitable classification of mechanical conditions can normally be carried out in accordance with [EN 50125-3].

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EL4-SSS-REQ-00774		Vibrations and Shocks - FES Components - Function		(Accepted)
Vibrations and shocks shall not impair the function of any [FES] component (Object controller, the cable system and all elements of the outdoor system) regarding the limit values of the housing in which the [FES] components are installed.				
The standard [EN 50125-3] - Chapter 4.13 (Vibration) shall be retained with the values [Outside the track (from 1m to 3m) from the rail].				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_184]	
Target Release:	DIL_OBB_1.0.x	See also:	==> [EL4-DIL-SSS-REQ-00816] Vibrations and Shocks - FES Components	
	DFE_1.0.0			
	DCA_OBB_3.1.0			
	AzLM_OBB_v7.10			
	IFC_1.0.0			
Verification:	Qualification test			
Issues/Decisions:				

3.1.1.3 PROTECTION RATING

EL4-SSS-REQ-00782		Protection Rating - Degree of Protection		(Accepted)
The components for use in the [FES] shall at least meet the requirements of [EN 60529] degree of protection IP20.				
Comments:				
Category:	ECR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_201]	
Target Release:	DIL_OBB_1.0.x	See also:		
	DFE_1.0.0			
	DCA_OBB_3.1.0			
	AzLM_OBB_v7.10			
	IFC_1.0.0			
Verification:	Qualification test			
Issues/Decisions:				

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3.1.2 MISCELLANEOUS

EL4-SSS-REQ-00738		OC - CPU usage		(In negotiation)
The [CPU] usage (computing power) of the [OC] shall not exceed 60% during initial commissioning in the customer network during normal operation.				
Comments: == OPEN POINT == Note: The [CPU] will exceed this limit when it is first admitted to the customer's network. In a further phase, more powerful [CPU]s with a maximum [CPU] workload of 60% will be integrated into the object controllers. Assumption of "normal operation": OC SW loaded. OC up and running. Field elements connected. No Interlocking commando via SCI available (point throwover, request for a train route, etc.).				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_102]	
Target Release:	DIL_OBB_2.0.0 DCA_OBB_3.1.2 AzLM_OBB_v7.11	See also:		
Verification:	Analysis, Module test			
Issues/Decisions:				

EL4-SSS-REQ-00739		OC - Reserve Capacity		(Accepted)
The supplier shall provide reserve computing power for [OC] enhancements (update/upgrade) to fulfill EL4-DIL-SSS-REQ-00738.				
The supplier shall inform the customer if enhancements to the [OC] exceed the reserve computing capacity.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_101]	
			==> [EL4 PH] [03_103]	
Target Release:	DIL_OBB_2.0.0	See also:	==> [EL4-DIL-SSS-REQ-00738] OC - CPU	
	DCA_OBB_3.1.2		usage	
	AzLM_OBB_v7.10			
Verification:	Analysis, Module test			
Issues/Decisions:				

EL4-SSS-REQ-00800		Software Update - Field Elements - Asynchronous Preloading		(Accepted)
All [EULYNX] field element subsystems shall implement the asynchronous preloading function according to [SMI].				
Comments:		SMI not yet specified by EULYNX. Therefore descoped from Phase 1. For DCA 3.1.0 Remote Update Function shall be delivered.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_268]	
Target Release:	DIL_OBB_3.0.0 DCA_OBB_3.1.2 AzLM_OBB_v7.11	See also:		
Verification:	System test			
Issues/Decisions:				

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3.2 RAMS

3.2.1 GENERAL

3.2.1.1 ASSUMPTIONS

Assumption 1:

No safety level is required for power supply and maintenance interfaces.

Assumption 2:

No safety level is required for the [DDSP].

Assumption 3:

No safety level is required for all security functions in the sense of safety.

Assumption 4:

Concerning the availability calculation for the manual repair of a [fault] an [MRT] of 0,5h is assumed.

Assumption 5:

Concerning the availability calculation for the manual repair of an [error] an [MRT] of 1h is assumed.

3.2.1.2 OEVE/OENORM AND EULYNX

EL4-SSS-REQ-00303		Fulfillment of Standard EN 50126		(Accepted)
The requirements according to [EN 50126] shall be fulfilled.				
Comments:	Compliance with the requirements of [EN 50126] is documented in the [RAM PLAN] and in the [SAFETY PLAN].			
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_23]	
Target Release:	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:		
Verification:	Analysis, Safety analysis			
Issues/Decisions:				

EL4-SSS-REQ-00306		Fulfillment of EULYNX Standard Eu.Doc.16		(Accepted)
The requirements according to [EULYNX Specification] - (Eu.Doc.16) shall be fulfilled.				
Comments:		<u>Note:</u> For the fulfillment of the requirements specified in [Eu.Doc.93] see the [SCI] specific requirements.		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_29]	
Target Release:	DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:		
Verification:	System test			
Issues/Decisions:				

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EL4-SSS-REQ-01478		Fulfillment of EULYNX Standard Eu.Doc.93		(Accepted)	
The requirements according to [EULYNX Specification] - (Eu.Doc.93) shall be fulfilled.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_29]		
Target Release:	DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:			
Verification:	System test				
Issues/Decisions:					

3.2.2 RELIABILITY

3.2.2.1 DEFINITIONS

Reliability:

Ability to function as required without [failure] under given conditions and for a given time interval (definition according to [EN 50126]).

[Fault]:

Refers to an incident that directly restrict system or operational performance and shall be resolved quickly.

[Error]:

Considered less critical for immediate operations, no operational impact.

[Failure]:

The state of an item in which it is unable to perform one or more of its required functions.

[Defect] classes:

→ FK1

Class 1: 'EMERGENCY'

[Fault]s that significantly hinder operations.

Example: Driving on sight/order, the DSTW being non-functional, or the failure of a central field element

→ FK2

Class 2: 'CRITICAL'

[Fault]s that have a significant impact on parts (functional/local) of the operation or serious [error]s.

Example: Driving on sight/order in a subsection of the DSTW, loss of redundancy that risks total failure, or failure of a field element affecting a partial area

→ FK3

Class 3 'MAJOR'

Other [fault]s or significant [error]s.

Example: Only **minor impact** on operation, or a serious error that has **no impact** on operation, or the failure of individual Virtual Interlocking (VSTW) functions.

→ FK4

Class 4 'MINOR'

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Other [error]s in the [DIL-CU] or outdoor system.

Example: **No operational impact**, an impact accepted by the Customer (AG), or failure of individual, less operationally relevant elements

3.2.2.2 IMPACT OF ERRORS AND FAULTS

EL4-SSS-REQ-00317		Reliability - Evidence by Calculation		(Accepted)	
The supplier shall provide an evidence by calculation of the required reliability in accordance with [CENELEC] specifications.					
Comments:		A mathematical proof of the required reliability will be provided to the customer in a RAM report in accordance with the specifications of a RAM plan and the requirements of the customer specifications.			
Category:	DOR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_37]		
Target Release:	DIL_OBB_1.0.x	See also:			
	DFE_1.0.0				
	DCA_OBB_3.1.0				
	AzLM_OBB_v7.10				
	IFC_1.0.0				
Verification:	Analysis				
Issues/Decisions:					

EL4-SSS-REQ-00321		Reliability - Error Reporting - DDSP		(Accepted)	
[Fault]s, [error]s and [failure]s of the [DIL] system shall be reported to the [DDSP].					
Comments:					
Category:		MDR		Priority:	
Originator:		Customer		Mandatory	
Target Release:		DIL_OBB_1.0.x		Origin:	
		DCA_OBB_3.1.0		==> [EL4 PH] [02_42]	
		AzLM_OBB_v7.10		See also:	
Verification:		System test			
Issues/Decisions:					

3.2.3 MAINTAINABILITY AND MAINTENANCE

3.2.3.1 DEFINITIONS

Maintainability:

The ability to remain in, or to be returned to, a functional state, under given application and maintenance conditions (definition according to [EN 50126]).

Maintenance:

Generic term for inspection, servicing and repair.

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3.2.3.2 GENERAL

EL4-SSS-REQ-00331 Maintainability and Maintenance - DDSP Compliance concerning Maintainability (Accepted)	
The DIL system shall comply with the [CRS_DDSP_Con] concept to ensure maintainability.	
Comments: In Version 3.1.0 remote update and SDI and SSI shall be supported.	
Category: DOR Originator: Customer Target Release: DIL_OBB_2.0.0 DCA_OBB_3.1.2 AzLM_OBB_v7.11	Priority: Mandatory Origin: ==> [EL4 PH] [02_53] See also:
Verification: System test	
Issues/Decisions:	

EL4-SSS-REQ-00332 Maintainability and Maintenance - IT Commissioning Compliance (Accepted)	
The DIL system shall comply with the [IT] commissioning concept to ensure maintainability.	
Comments: This refers to the [LH], part D9 "IT-Inbetriebnahme Konzept" [CRS_IT_PiO], and part D8 "Security Konzept" [CRS_SEC_CONCEPT].	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	Priority: Mandatory Origin: ==> [EL4 PH] [02_54] See also:
Verification: System test	
Issues/Decisions:	

3.2.3.3 STORAGE OF SPARE PARTS

EL4-SSS-REQ-00337 Maintainability and Maintenance - Spare Parts Storage - Maintenance (Accepted)	
All spare parts shall be storable without requiring regular maintenance and without being kept under voltage.	
Comments: Exceptions for individual components require an alternative solution from the contractor and shall be approved by the customer.	
Category: MDR Originator: Customer Target Release: DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	Priority: Mandatory Origin: ==> [EL4 PH] [02_59] See also:
Verification: System test	
Issues/Decisions:	

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3.2.3.4 LABELLING

EL4-SSS-REQ-00339		Maintainability and Maintenance - Labelling - Data	(In negotiation)
All hardware components and modules of the indoor and outdoor systems, including switch cabinets and switch boxes, shall be clearly labelled in machine-readable form using a [QR] code and shall show at least the following data (if available): → Manufacturer → Type → Item number → Device version (hardware and firmware version, etc.) → Serial number → [MAC] addresses			
<i>Comments:</i> Hitachi Rail intends to apply the GS1 data matrix code (2D barcode) instead of the QR code. Project specific label to be defined with customer first.			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_62]
<i>Target Release:</i>	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>See also:</i>	
<i>Verification:</i>	System test		
<i>Issues/Decisions:</i>			

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EL4-SSS-REQ-00340		Maintainability and Maintenance - Labelling - Units	(In negotiation)
Labelling shall be applied to the smallest replaceable unit, for example: → Object Controller → Power supply units → Light point → Wheel sensor → Lightning protection → Signal box → Separately mountable parts by the supplier → [FES] → [IFC Block - SCA] → [IFC Block - BSAU] → [IFC Block - IOC] → Server			
<i>Comments:</i>			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_63]
<i>Target Release:</i>	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>See also:</i>	
<i>Verification:</i>	System test		
<i>Issues/Decisions:</i>			

EL4-SSS-REQ-00343		Maintainability and Maintenance - Labelling - Serial Number	(Accepted)
The serial number of the components in the spare parts inventory shall enable a unique identification of the components throughout their entire operating life.			
<i>Comments:</i> The serial numbers of the components allow for transparent tracking of components throughout their entire operating life (e.g. in [DDSP], [AVS], [SAP]).			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_66] ==> [EL4 PH] [02_67]
<i>Target Release:</i>	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>See also:</i>	
<i>Verification:</i>	System test		
<i>Issues/Decisions:</i>			

3.2.3.5 MAINTENANCE DOCUMENTATION

Maintenance is carried out in accordance with the service description and service level agreement.

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EL4-SSS-REQ-00345 Maintainability and Maintenance - Documentation - Usability (Accepted)	
The supplier shall deliver the maintenance documentation in a form that enables the customer to perform the inspection, maintenance and repair (at least component replacement, troubleshooting, resetting, restarting, etc.) of the supplied system.	
<i>Comments:</i>	
<i>Category:</i> DOR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_70] <i>See also:</i>
<i>Verification:</i> Review	
<i>Issues/Decisions:</i>	

3.2.3.6 COMPONENT REPLACEMENT

EL4-SSS-REQ-00353 Maintainability and Maintenance - Component Replacement - OC Replacement - Restrictions (Accepted)	
The [OC]s shall be designed in such a way that replacement with an incorrect spare part is not possible.	
<i>Comments:</i> If it is not possible to distinguish between different [OC]s using hardware, the wrong [OC] shall be detected by software. A status error with [LED] must be displayed on the device.	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_80] <i>See also:</i>
<i>Verification:</i> System test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00356 Maintainability and Maintenance - Component Replacement - Standard Tools (Accepted)	
No specialized tool (designed to work only with the component being replaced) shall be required to replace any component.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_83] <i>See also:</i>
<i>Verification:</i> Review	
<i>Issues/Decisions:</i>	

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3.2.3.7 EMPLOYEE PROTECTION

EL4-SSS-REQ-00566		Maintainability and Maintenance - Employee Protection - Product Safety		(Accepted)
The supplier shall comply with the requirements concerning product safety according to [EN 50122] and [EN 61010-1] to ensure employee protection.				
Components, subsystems and systems supplied by the supplier shall be designed in such a way that the employee protection is ensured when the customer carries out maintenance works.				
Comments:		The employee protection for the customer's employees can only be ensured by the customer himself.		
		For network components, the standard [EN 62368-1] shall also be applied.		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_92]	
Target Release:	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:	==> [EL4-DIL-SSS-REQ-00785] Employee Protection - Preventive Services/Forces	
Verification:	Review			
Issues/Decisions:				

3.2.4 AVAILABILITY

3.2.4.1 DEFINITIONS

Availability:

The ability to perform a required function at a given time or during a given time interval under given conditions, provided that the necessary external aids are provided (definition according to [EN 50126]).

Automatic correction:

In the context of a "safe computing unit" (e.g. a [VIL Node]) means the following:

After a [fault] has been detected, a safe computing unit enters a safe state and performs a self-test. After a successful self-test, the disturbed module is automatically recommissioned.

This also applies to the other availability requirements in this chapter.

3.2.4.2 AVAILABILITY CLASSES

Depending on the required availability values, the following availability classes are defined:

- Class 1: Availability is 99.999 %
- Class 2: Availability is 99.995 %
- Class 3: Availability is 99.99 %

The availability values were determined in accordance with [EN 50126] in a top-down approach as follows:

$$A = [MTTF]/([MTTF]+[MTTR])$$

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3.2.4.3 OC AVAILABILITY

EL4-SSS-REQ-00581		Availability - OC - Overall		(Accepted)
The [OC]s shall have a class 3 overall availability of at least 99.99 %.				
Comments: [OC] refers to [LS], [P], [TDS] and [Gen-IO] This requirement (Class 3) defines the overall availability target of the Object Controller (OC) throughout one operating year. The value 99.99% corresponds to a maximum downtime of approximately 52.6 minutes per year. The overall availability of 99.99% is a minimum requirement for the entire OC across all fault types.				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_132]	
Target Release:	DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:	==> [EL4 PH] [02_134]	
Verification:	Analysis			
Issues/Decisions:				

EL4-SSS-REQ-00582		Availability distribution of [OC]		(Accepted)
The distribution of the availability of a [OC] shall correspond to class 2. This means in particular an availability of				
99.9957 % (MTTR = 0.5 h)				
99.999 % (MTTR = 10 min)				
99.9996 % (MTTR = 2 min)				
<i>Comments:</i> [OC] refers to [LS], [P], [TDS] and [Gen-IO]				
Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.				
For manual repair, an MRT = 0.5 h is assumed.				
<i>Category:</i>	DPR	<i>Priority:</i>	Mandatory	
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_135]	
			==> [EL4 PH] [02_137]	
			==> [EL4 PH] [02_138]	
			==> [EL4 PH] [02_195]	
<i>Target Release:</i>	DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>See also:</i>		
<i>Verification:</i>	Analysis			
<i>Issues/Decisions:</i>				

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3.2.5 SAFETY

3.2.5.1 TOLERABLE HAZARD RATES

EL4-SSS-REQ-00592		Safety - THR - IO		(Accepted)			
The [THR] value of the subsystem [IO] shall be less or equal 1.0E-9 1/h.							
Comments:							
Category:		SAR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_166]	
Target Release:		DIL_OBB_1.0.x		See also:			
		DFE_1.0.0					
		DCA_OBB_3.1.0					
Verification:		Safety analysis					
Issues/Decisions:							

EL4-SSS-REQ-00684		Safety - Integrity Levels - Interface - OC		(Accepted)			
The interfaces of the [VIL] [SCI-LS], [SCI-TDS], [SCI-P], [SCI-IO] to the [OC]s light signal, track detection system, points, generic IO shall fulfil the requirements according to Safety Integrity Level 4 [SIL]4.							
Comments:							
Category:		SAR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_172]	
Target Release:		DIL_OBB_1.0.x		See also:			
		DFE_1.0.0					
		DCA_OBB_3.1.0					
		AzLM_OBB_v7.10					
Verification:						Safety analysis	
Issues/Decisions:							

EL4-SSS-REQ-00686		Safety - Integrity Levels - Development DIL		(Accepted)	
The development of the [DIL] shall be carried out in accordance with the safety requirement level [SIL]4 and the procedures and techniques requested in [EN 50129] shall be taken into account.					
Comments:		Compliance with the requirements of [EN 50129] is documented in the Safety Plan.			
Category:	SAR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_175]		
Target Release:	DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:	==> [EL4-DIL-SSS-REQ-00305] Fulfillment of Standard EN 50129		
Verification:	Safety analysis				
Issues/Decisions:					

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EL4-SSS-REQ-00687 Safety - Integrity Levels - SW VIL, OC (Accepted)	
The software of the [VIL] and the [OC]s shall be developed in accordance with the specified software safety requirements and the procedures and techniques requested in [EN 50716] for it.	
<i>Comments:</i> OEVE EN 50716 was published on October 1, 2024, specifying the withdrawal of EN 50128 at the end of October 2026. Therefore, EN 50716 shall be used for the DIL. Compliance with the requirements of [EN 50716] is documented in the [SAFETY_PLAN].	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_176] <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00304] Fulfillment of Standard EN 50716
<i>Verification:</i> Safety analysis	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00689 Safety - Integrity Levels - Data Communication (Draft)	
For data communication via open (categories 2 and 3) and closed (category 1) networks, [EN 50159] shall be complied with. The OCs shall provide mechanism for secure data communication over category 3 networks.	
<i>Comments:</i> Definition of Classes for Network required. SiDiNet Category 3 X25oRaSTA Category 2	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_178] <i>See also:</i>
<i>Verification:</i> Safety analysis	
<i>Issues/Decisions:</i>	

3.2.5.2 MISCELLANEOUS

EL4-SSS-REQ-00694 Safety - Power Supply - Shutdown (Accepted)	
In case of a power [failure], the [VIL] and all components of the outdoor system shall enter a safe state.	
<i>Comments:</i> A power failure leads to a disruption of communication with the affected system. The higher-level communication partner must ensure that no unsafe state arises for all affected subsystems as a whole.	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_39] ==> [EL4 PH] [02_184] <i>See also:</i>
<i>Verification:</i> System test	
<i>Issues/Decisions:</i>	

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EL4-SSS-REQ-00412 Safety - Data Configuration Process (Accepted)	
The supplier shall provide a certified [CENELEC]-compliant data configuration process for the [VIL] subsystem, the [OC] subsystems and all interfaces to the sub-systems and peripheral systems.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_188] <i>See also:</i>
<i>Verification:</i> Analysis	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00413 Safety - Safety Case (Accepted)	
The validity of the safety case shall not be limited in time.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_190] <i>See also:</i>
<i>Verification:</i> Review	
<i>Issues/Decisions:</i>	

3.3 NETWORKS

3.3.1 SEPARATION OF INTERFACES

EL4-SSS-REQ-00543 Networks - Interfaces Redundant No VLAN Separation per EULYNX (Draft)	
All interfaces ([SCI], [SMI], [SDI] and [SSI]) shall be realized with redundancy (two physical [PoS] signaling interfaces). The [SCI], [SMI], [SDI] and [SSI] interfaces are not separated based on [VLAN] (see [EULYNX Specification] - (Eu.Pos.1143)).	
<i>Comments:</i> - Both [OS] and the [DIL-CU] servers are equipped with several network interfaces. - The PoS signaling interface is the switch port on the first network device that is part of the [SiDiNet] to which the cable of the EULYNX system or an adjacent system is connected (see [Eu.Pos.980]).	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_18] ==> [EL4 PH] [22_19] ==> [EL4 PH] [22_20] <i>See also:</i>
<i>Verification:</i> System test	
<i>Issues/Decisions:</i>	

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EL4-SSS-REQ-00544 Networks - Interface Traffic Prioritized via DSCP (Draft)	
per EULYNX	
The prioritization [QoS] of the network traffic of the [SCI], [SMI], [SDI] and [SSI] interfaces shall be done using [DSCP] markers according to [EULYNX Specification] - (Eu.Pos.1140).	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_21] <i>See also:</i>
<i>Verification:</i> System test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00545 Networks - Sender Marks Interface Traffic with (Draft)	
DSCP	
The traffic on the [SCI], [SMI], [SDI] and [SSI] interfaces shall be provided with a [DSCP] traffic marker by the sender.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_22] <i>See also:</i>
<i>Verification:</i> System test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00546 Networks - Separate Networks shall Use Distinct (Draft)	
IP Segments	
The physically separate networks with their physically separate [PoS] signaling interfaces shall use different IP segments.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_23] <i>See also:</i>
<i>Verification:</i> Module test	
<i>Issues/Decisions:</i>	

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EL4-SSS-REQ-00547 Networks - NTP Traffic Marked per EULYNX/EU-Rail SP Interface Spec (Draft)	
The [NTP] traffic to/from the [ÖBB] time server shall be marked with the traffic marker for the interface specified by EULYNX or EU-Rail SP documents.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_25] <i>See also:</i>
<i>Verification:</i> Module test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00548 Networks - Customer-Defined DSCP Markers Must Be Configurable (Draft)	
The respective [DSCP] traffic markers (set by the customer) shall be configurable.	
<i>Comments:</i> • The [DSCP] traffic markers are defined during the realization. • The [QoS] priorities in the [SiDiNet] are set according to the [PoS] signalling.	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_26] ==> [EL4 PH] [22_27] ==> [EL4 PH] [22_29] <i>See also:</i>
<i>Verification:</i> Review	
<i>Issues/Decisions:</i>	

3.3.2 PROTOCOL STACK

EL4-SSS-REQ-00549 Networks - IPv4 Required at OSI Layer 3 per EULYNX (Draft)	
The network layer ([OSI] layer 3) shall use [IPv4] (see [EULYNX Specification] - (Eu.Pos.1107)).	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [22_31] <i>See also:</i>
<i>Verification:</i> Review	
<i>Issues/Decisions:</i>	

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EL4-SSS-REQ-00550		Networks - No Port Channels or Link Aggregation Protocols		(In negotiation)	
No port channels or link aggregation protocols shall be used between the two [PoS] signaling interfaces (see [EULYNX Specification] - (Eu.Doc.15).					
Comments: == OPEN POINT ==					
The [DIL-CU] servers and the [OC]s do not use port channels or link aggregation protocols for the [PoS] signaling interfaces. However, from an architectural point of view, these protocols are necessary for the other interfaces (e.g. data exchange between [ADM], [VIL] instances)					
Reference Eu.Sec.363 in Eu.Doc.15 had been removed with EULYNX BL4R2. Unclear if customer still require this to be covered.					
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [22_32]		
Target Release:	DIL_OBB_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:			
Verification:	Review				
Issues/Decisions:					

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4. COMMON REQUIREMENTS FOR EULYNX SUBSYSTEM IO

4.1 GENERAL REQUIREMENTS

4.1.1 ENVIRONMENT AND PROTECTION

EL4-DFE-SSS-REQ-00069 Environmental Conditions - EN 50125-3 Temperature Range (Accepted)	
As the components within the [FES] shall be operated indoor as well as in a cubicle next to the track, for operation the temperature range of -40 °C to +70 °C shall be maintained (extends class T2 in [EN 50125-3]).	
Comments: Ambient temperature outside the [FES] (outdoor): -40 °C to +45 °C	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00596] Environmental Conditions - DIL - Climate Class T2 (Indoors and Outdoors)
Target Release: DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:
Verification: Qualification test	
Issues/Decisions:	

EL4-DFE-SSS-REQ-00070 Environmental Conditions - EN 50125-3 Humidity Range (Accepted)	
For operation, the humidity range of 5 % to 100 % (0.12 g/m ³ to 30 g/m ³ (tunnel)) shall be maintained for components within the [FES].	
Comments:	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00746] Environmental Conditions - FES - Climate Class T2
Target Release: DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:
Verification: Qualification test	
Issues/Decisions:	

EL4-DFE-SSS-REQ-00068 Environmental Conditions - EN 50125-3 Altitude Class (Accepted)	
Altitude class AX in accordance with [EN 50125-3] - up to 2000 m - shall be taken into account for components within the [FES].	
Comments: Stricter requirement: (2000m)	
Category: FUR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00595] Environmental Conditions - FES, OC - Altitude Class A1
Target Release: DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:
Verification: Module test, Qualification test, Subit test	
Issues/Decisions:	

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EL4-DFE-SSS-REQ-00018 Storage Conditions		(Accepted)
The components within the [FES] shall be suitable for storage in their packaging under the following conditions as specified in [EN 60721-3-1]. However, with an extended upper ambient temperature of +70 °C:		
• Climatic Conditions: class 1K22 • Resistance against: • Biologic influences: class 1B1 • Chemically active substances: class 1C2 • Mechanically active substances: class 1S11 • Mechanical environmental conditions: 1M11 (Test duration: 120 minutes in each axis direction (see [EN 60068-2-64]))		
Comments:		
Category: MDR Originator: Customer	Priority: Mandatory Origin: ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00334] Maintainability and Maintenance - Spare Parts Storage - Concept ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00335] Maintainability and Maintenance - Spare Parts Storage - Information ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00338] Maintainability and Maintenance - Spare Parts Storage - Climate Class	
Target Release: DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:	
Verification: Qualification test		
Issues/Decisions:		

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EL4-DFE-SSS-REQ-00480		RoHS		(Draft)	
When selecting components, the evaluation and restriction of hazardous chemicals under [RoHS] in accordance with Directive 2011/65/EU shall be taken into account.					
Comments:					
Category:		PER		Priority:	
Originator:		Customer		Mandatory	
Target Release:		DCA_OBB_3.1.0		Origin:	
		AzLM_OBB_v7.10		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00778]	
		IFC 1.0.0		Environmental Protection - Materials in	
				Case of Fire	
Verification:		Qualification test		See also:	
Issues/Decisions:					

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EL4-DFE-SSS-REQ-00192		RoHS		(Draft)	
All circuit boards shall be compliant to the restriction of the use of certain hazardous substances in electrical and electronic equipment [RoHS] according the Directive 2011/65/EU.					
Comments:					
Category:	PER	Priority:	Mandatory	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00604] EMS - DC Server ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00778] Environmental Protection - Materials in Case of Fire	
Originator:	Customer	Origin:			
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:			
Verification:	Qualification test				
Issues/Decisions:					

EL4-DFE-SSS-REQ-00406		REACH		(Draft)	
When selecting components, the evaluation and restriction of hazardous chemicals under [REACH] in accordance with Directive 1907/2006 shall be taken into account.					
Comments:					
Category:	PER	Priority:	Mandatory	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00778] Environmental Protection - Materials in Case of Fire	
Originator:	Customer	Origin:			
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:			
Verification:	Qualification test				
Issues/Decisions:					

EL4-DFE-SSS-REQ-00408		Case of fire		(Draft)
In case of fire corrosive and toxic fumes shall be avoided. Materials shall be halogen-free according to [EN 60754-1].				
Comments:		[EN 60754-1] - Test on gases evolved during combustion of materials from cables - Part 1: Determination of the halogen acid gas content		
Category:	PER	Priority:	Mandatory	
Originator:	Product Management	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00778] Environmental Protection - Materials in Case of Fire	
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:		
Verification:	Review			
Issues/Decisions:				

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4.1.2 HARDWARE - MISCELLANEOUS

EL4-DFE-SSS-REQ-00196		Plug Marker		(Draft)	
All plugs and sockets in replaceable components shall be marked unambiguously.					
Comments:					
Category:	CSR		Priority:	Mandatory	
Originator:	Product Management		Origin:		
Target Release:	DFE_1.0.0		See also:		
	DCA_OBB_3.1.0				
	AzLM_OBB_v7.10				
	IFC_1.0.0				
Verification:	Subit test				
Issues/Decisions:					

4.1.3 EULYNX IMPLEMENTATION

EL4-DFE-SSS-REQ-00083		Proof of Compliance		(Accepted)	
The DCA system shall show compliance to [EULYNX Specification], to referenced specifications and to System Requirements Specifications of the customer project by providing a validation report.					
Comments:		The handover of a RCM (Requirements Conformity Matrix) without validation is insufficient.			
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:			
Target Release:	DFE_1.0.0	See also:			
	DCA_OBB_3.1.0				
	AzLM_OBB_v7.10				
Verification:	Subit test				
Issues/Decisions:					

EL4-DFE-SSS-REQ-00060		General - Import of Interlocking Station Data		(Accepted)	
The OC Data Preparation system shall be able to import previous defined handover files from the Interlocking Data Preparation system.					
Comments:					
Category:		FUR		Priority:	
Originator:		Technics (R&D)		Mandatory	
Target Release:		DCA_OBB_3.1.0		Origin:	
		AzLM_OBB_v7.10		See also:	
Verification:		Module test, Subit test			
Issues/Decisions:					

EL4-DFE-SSS-REQ-00061		General - Data Preparation		(Accepted)	
The DCA system shall provide a proper data preparation and build environment.					
Comments:					
Category:		FUR		Priority:	
Originator:		Technics (R&D)		Mandatory	
Target Release:		DCA_OBB_3.1.0		Origin:	
		AzLM_OBB_v7.10		See also:	
Verification:		Module test, Subit test			
Issues/Decisions:					

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EL4-DFE-SSS-REQ-00066		Automatic Self-Test		(Accepted)	
The DCA system shall be able to perform an automatic self-test on request by the DIL system. The results of the self-test shall be a standardised report on the status of the OC to the DDSP.					
Comments:		current DCA checks include: - HW online tests (OLT) implemented by TAS PLF, namely: - o Disk test (storage media like flash) - o Memory tests: controller, application in memory - o Hardware watchdog, - o Timelines: ensure that a real time application becomes idle within a specified time. - o Clock and timer drifts - o CPU resources (speed, number of cores, amount of RAM memory) - o Environment (validation of static data, check if all tests are active, ..) - Failures in inputs, outputs, disconnection relays - SW watchdog			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00740] Automatic Self Test - Request from DDSP		
Target Release:	DCA_OBB_3.1.2 AzLM_OBB_v7.11	See also:			
Verification:	Module test, Subit test				
Issues/Decisions:					

4.1.4 OBJECT CONTROLLER

4.1.4.1 CAN - INTERFACE - DEVICES

EL4-DFE-SSS-REQ-00147		Automatic Self-Test with external CAN-Devices		(Accepted)	
DCA subsystems connected via CAN bus (e.g., iLED) shall be checked during the automatic self-test. This includes checking for connection errors, malfunctions, and limitations of these external components.					
Comments:					
Category:		FUR		Priority:	
Originator:		Customer		Mandatory	
				Origin:	
				==> [EL4 SSS] [EL4-DIL-SSS-REQ-00740]	
Target Release:		DCA_OBB_3.1.2		Automatic Self Test - Request from DDSP	
				See also:	
Verification:		Module test, Subit test			
Issues/Decisions:					

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EL4-DFE-SSS-REQ-00149		Object Controller - SW-Update for external CAN-Devices		(Accepted)	
A software update for DCA subsystems connected via CAN bus (e.g. iLED) shall be supported.					
Comments:					
Category:		FUR		Priority:	
Originator:		Customer		Mandatory	
				Origin:	
				==> [EL4 SSS] [EL4-DIL-SSS-REQ-00739]	
Target Release:		DCA_OBB_3.1.0		OC - Reserve Capacity	
Verification:		See also:			
		Module test, Subit test			
Issues/Decisions:					

4.1.4.2 BASIC DATA IDENTIFIER

The parameters required in the Basic Data Identifier for certificate issuance are defined in the [CRS_Sec].

EL4-DFE-SSS-REQ-00410		Basic Data Identifier - uniquely and unambiguously		(Draft)
Each object controller shall be parameterized uniquely and unambiguously with a Basic Data Identifier in accordance with [EULYNX] requirements.				
Comments:				
Category:	CSR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00894] Interfaces - P4 - Basic Data Identifier ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00903] Interfaces - LS6 - Basic Data Identifier ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00523] Interfaces - IO1 - Basic Data Identifier ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00536] Interfaces - TDS1 - Basic Data Identifier ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00794] EULYNX - Basic Data Identifier - Compliance	
Target Release:	DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:		
Verification:	Subit test			
Issues/Decisions:				

EL4-DFE-SSS-REQ-00430		Basic Data Identifier - physically identical		(Draft)			
The Basic Data Identifier shall be physically identical for all EULYNX subsystems and Object Controller types.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00795] EULYNX - Basic Data Identifier - Subsystem	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10		See also:			
Verification:		Subit test					
Issues/Decisions:							

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EL4-DFE-SSS-REQ-00431 Basic Data Identifier - Security (Draft)	
The Basic Data Identifier shall fulfil the security requirements according to the [SSS_DIL_SECURITY].	
<i>Comments:</i> == OPEN POINT == The security requirements regarding Basic Data Identifier will be specified in more detail during Security Threat Assessment.	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00796] EULYNX - Basic Data Identifier - Security <i>See also:</i>
<i>Verification:</i> Subit test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00432 Basic Data Identifier - Automatic Initialization (Draft)	
The Basic Data Identifier shall contain all data necessary for automatic initialization of an DCA Object Controller.	
The configuration shall contain at least:	
• Network configuration (IP address, subnet mask, gateway, DNS server) • Technical EULYNX ID • Address or host name of the DDSP • Address or host name of the PKI	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00797] EULYNX - Basic Data Identifier - Initialisation <i>See also:</i>
<i>Verification:</i> Subit test	
<i>Issues/Decisions:</i>	

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EL4-DFE-SSS-REQ-00433		Basic Data Identifier - Unique initial Configuration		(Draft)
The unique initial configuration (identifier, initial encryption) shall also be retained when the Object Controller is replaced. All location specific data shall be derived from the unique initial configuration. Spare parts shall be exchangeable between locations of the same Object Controller type.				
<i>Comments:</i>				
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory	
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00798] EULYNX - Basic Data Identifier - Retainment	
<i>Target Release:</i>	DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>See also:</i>		
<i>Verification:</i>	Subit test			
<i>Issues/Decisions:</i>				

EL4-DFE-SSS-REQ-00434	Basic Data Identifier - Protection against unintentional Change		(Draft)
The initial configuration of the Basic Data Identifier shall be protected against unintentional change or data loss.			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00799] EULYNX - Basic Data Identifier - Non-Volatility
<i>Target Release:</i>	DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>See also:</i>	
<i>Verification:</i>	Review, Security test		
<i>Issues/Decisions:</i>			

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4.2 SUBSYSTEM FIELD ELEMENT STATION

4.2.1 GENERAL

EL4-DFE-SSS-REQ-00105		Object controller power supply protection		(In negotiation)	
The power supply input for each object controller shall be protected separately.					
Comments:		OPEN POINT: This requirement is not fulfilled in case several OC share the same backplane. Requires customer approval.			
Category:	CSR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00893] Interfaces - P2 - Power Supply ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00912] Interfaces - LS8 - Power Supply ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00525] Interfaces - IO4 - Power Supply ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01061] Power Supply - Electrical Protection for Dis- /Connections		
Target Release:	DFE_1.0.0 DCA_OBB_3.1.2 AzLM_OBB_v7.10	See also:			
Verification:		Subit test			
Issues/Decisions:					

EL4-DFE-SSS-REQ-00439		Flammability of cables		(Draft)	
Cables shall be flame-resistant according to [EN60332-1-2] and [EN60332-2-2].					
Comments:		EN 60332 - Tests on electric and fibre optic cables under fire conditions.			
Category:	PER	Priority:	Mandatory		
Originator:	Product Management	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00786] Employee Protection - Fire Protection - General ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01046] Network - Laying and Connecting Fibre Optic Cables - Regulations 2		
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:			
Verification:	Subit test				
Issues/Decisions:					

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4.3 RAMS

4.3.1 GENERAL

EL4-DFE-SSS-REQ-00003 RAMS - Supply of Spare Parts - FES		(Accepted)
The components within the [FES] shall ensure the supply of all spare parts to maintain operation and full functionality for the minimum usage period of 25 years.		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00302] Supply for Spare Parts ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01011] Minimum Useful Life - Outdoor System Components
<i>Target Release:</i>	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>See also:</i>
<i>Verification:</i>	Analysis	
<i>Issues/Decisions:</i>		

EL4-DFE-SSS-REQ-00002 RAMS - Minimum Usage Extension - FES		(Accepted)
The minimum usage period of components within the [FES] might be extended once by 5 years.		
<i>Comments:</i> The FRACAS system is used to examine the reliability of the DSTW components of the outdoor system.		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00301] Minimum Useful Life - Extension
<i>Target Release:</i>	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>See also:</i>
<i>Verification:</i>	Subit test	
<i>Issues/Decisions:</i>		

4.3.2 MAINTAINABILITY

4.3.2.1 COMPONENT REPLACEMENT

The end customer provides a data diagnosis service platform (DDSP) as a software tool for maintenance. The DDSP system represents the subsystems MDM and SSP according to Eulynx reference architecture.

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EL4-DFE-SSS-REQ-00022 RAMS - Reconfiguration/Modification of Data Preparation (Accepted)	
It shall not be possible for maintenance personnel on site to reconfigure or modify the data preparation of an Object Controller when replacing one Object Controller due to faults or malfunctions.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00352] Maintainability and Maintenance - Component Replacement - Restrictions <i>See also:</i>
<i>Verification:</i> Analysis	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00028 RAMS - Max. Boot Time of OC (Accepted)	
The Maximum Operational Startup Time of the Object Controller until it reaches full operational readiness shall not exceed 180 seconds. This duration includes the full system boot, the successful connection and recognition of all CAN-bus devices, and if necessary the time required for activating a downloaded Software Release.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> <i>See also:</i>
<i>Verification:</i> Analysis, Performance test, Subit test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00011 RAMS - Withstand Current Peaks (Accepted)	
Power contacts between back panel and Object Controller shall withstand current peaks during hot plugging greater than 30 times.	
<i>Comments:</i> Caution: ELCAP at first contact	
<i>Category:</i> DPR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00313] Availability - LRU - No Maintenance Impact <i>See also:</i>
<i>Verification:</i> Analysis, System test	
<i>Issues/Decisions:</i>	

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4.3.2.2 MAINTENANCE-FREE

EL4-DFE-SSS-REQ-00030		RAMS - Maintenance-Freeness - FES		(Accepted)	
Field element stations and components within the [FES] shall be maintenance-free.					
Comments:		If maintenance is required, it shall be limited to once of 6 months and shall be documented (e.g. in the maintenance manual).			
Category:	MDR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00360] Maintainability and Maintenance - No preventive Maintenance ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01479] Maintainability and Maintenance - No Inspection and Maintenance		
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	See also:			
Verification:	Review				
Issues/Decisions:					

4.3.2.3 EMPLOYEE PROTECTION

EL4-DFE-SSS-REQ-00031		RAMS - Housings - No Edges		(Accepted)	
The housing including the heat sink shall not have any sharp edges.					
Comments:					
Category:		GCR	Priority:		Mandatory
Originator:		Customer	Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:		
Verification:		Review, Module test			
Issues/Decisions:					

EL4-DFE-SSS-REQ-00032		RAMS - Housings - Surface Temperature		(Accepted)			
The housing surface temperature shall not exceed 65°C at any point up to an ambient temperature of 40°C. For higher ambient temperatures, the maximum difference is 25°C. [EN61010-1 - Table 19]							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Technics (R&D)		Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0		See also:			
Verification:		Qualification test					
Issues/Decisions:							

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EL4-DFE-SSS-REQ-00033		RAMS - Markings		(Accepted)	
Required markings shall remain clear and legible under conditions of normal use and shall resist the effects of the cleaning agents specified by the manufacturer.					
After a conformity check (see "Comments") the markings shall be clearly legible and adhesive labels shall not have worked loose or become curled at the edges.					
<i>Comments:</i> <u>Conformity check:</u> The markings are rubbed by hand, without undue pressure, for 30 s with a cloth soaked with each specified cleaning agent (or, if not specified, with 70 % isopropyl alcohol).					
<i>Category:</i> GCR		<i>Priority:</i> Mandatory		<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety	
<i>Originator:</i> Technics (R&D)		<i>Origin:</i>			
<i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0		<i>See also:</i>			
<i>Verification:</i> Qualification test					
<i>Issues/Decisions:</i>					

EL4-DFE-SSS-REQ-00034		RAMS - Connections to external Circuits		(Accepted)			
Connections to external circuits shall not, under normal conditions or under the conditions of a single fault in the device, result in accessible parts of the device or accessible parts of an external circuit resulting in danger.							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Technics (R&D)		Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0		See also:			
Verification:		Analysis					
Issues/Decisions:							

EL4-DFE-SSS-REQ-00035		RAMS - Accessible conductive Parts		(Accepted)	
Accessible conductive parts of terminals that carry voltage due to a charge of internal capacitors shall not have a dangerous voltage/charge for 10 seconds after the power supply has been interrupted.					
Plug pins of devices with mains connection cables shall not have any dangerous voltage/charge even after 5 seconds after the plug connection has been removed.					
Comments:					
Category: GCR		Priority: Mandatory		See also:	
Originator: Technics (R&D)		Origin: ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety			
Target Release: DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0					
Verification: Analysis, Integration test					
Issues/Decisions:					

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EL4-DFE-SSS-REQ-00036		RAMS - Insulation Coordination		(Accepted)	
When defining insulation coordination, in addition to the railway standard [EN 50124-1], the requirements of the employee protection standard [EN 61010-1] shall be fulfilled for components within the [FES].					
Comments:					
Category:		GCR		Priority:	
Originator:		Technics (R&D)		Mandatory	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0		Origin:	
Verification:		Review, Qualification test		See also:	
Issues/Decisions:				==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety	

EL4-DFE-SSS-REQ-00037		RAMS - Fire Protection		(Accepted)	
Regarding fire protection, the standard [EN61010-1] - Chapter 9 shall be observed for components within the [FES].					
Comments:		For network components, the standard [EN62368-1] shall also be applied.			
Category:	GCR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety		
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	See also:			
Verification:	Review				
Issues/Decisions:					

EL4-DFE-SSS-REQ-00038		RAMS - Part holding Terminals		(Accepted)			
If terminals conduct a current of more than 0.5 A during normal operation and can dissipate considerable heat in the event of insufficient contact, the part holding the terminals shall be made of an insulating material that does not soften to the point where it could cause a hazard or a further short circuit. [EN 61010-1] - Chapter 10.5.3.							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Technics (R&D)		Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00566] Maintainability and Maintenance - Employee Protection - Product Safety	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0		See also:			
Verification:		Analysis					
Issues/Decisions:							

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4.3.3 AVAILABILITY

4.3.3.1 MONITORING SYSTEM [FRACAS]

EL4-DFE-SSS-REQ-00213 FRACAS - Automatic recovery of FES (Draft)	
A [FES] (incl. OCs) and block converter shall support automatic recovery after power restoration without manual intervention.	
<i>Comments:</i> affects MTTR and therefore Availability	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>Priority:</i> Mandatory <i>Origin:</i> <i>See also:</i>
<i>Verification:</i> Integration test, System test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00214 FRACAS - OC Failure Isolation within FES (Draft)	
Architecture	
A failure of one Object Controller (OC) must not affect the operation of other OC's within the same [FES].	
<i>Comments:</i>	
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> <i>See also:</i>
<i>Verification:</i> Review, Module test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00215 FRACAS - Impact-Free Replacement of OC (Draft)	
Components	
The replacement of an OC shall not result in any operational effects or [fault]s other than the replacement of this OC.	
<i>Comments:</i> addressed in RAM HW-FMEA	
EL4-DIL-SSS-REQ-00354	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00313] Availability - LRU - No Maintenance Impact <i>See also:</i>
<i>Verification:</i> Integration test, System test	
<i>Issues/Decisions:</i>	

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EL4-DFE-SSS-REQ-00401 FRACAS - MRT - Component replacement (event of fault) (Draft)			
In the event of a fault, the required MRT value of a maximum of 0.5 h shall be demonstrably adhered to for the replacement of components during maintenance.			
<i>Comments:</i>		The startup time of the OC must be taken into account.	
<i>Category:</i>	CSR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00357] Maintainability and Maintenance - Component Replacement - Manual corrective Maintenance Actions
<i>Target Release:</i>	DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10 IFC_1.0.0	<i>See also:</i>	
<i>Verification:</i>	Subit test		
<i>Issues/Decisions:</i>			

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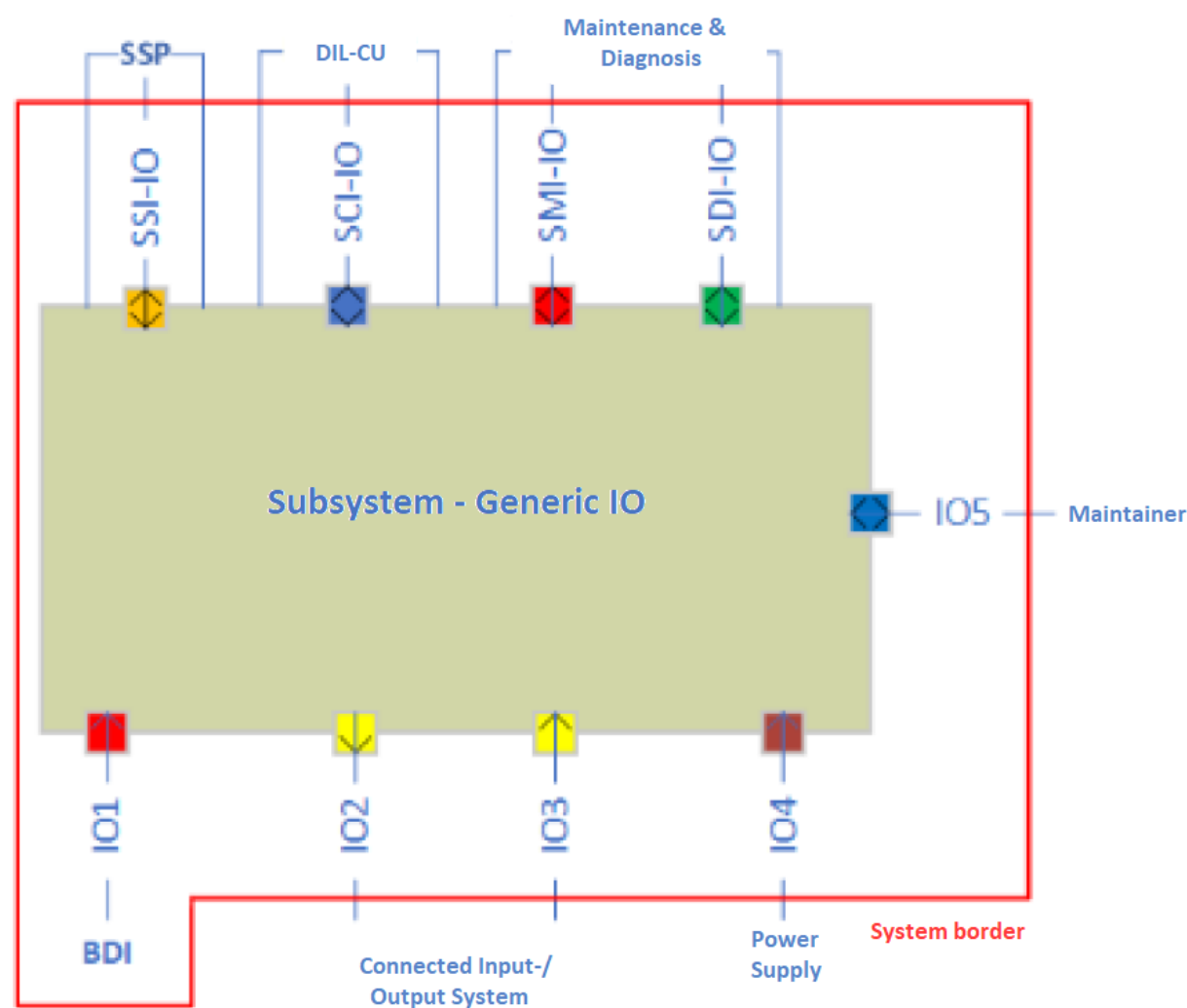
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5. SUBSYSTEM REQUIREMENTS FROM SYSTEM SPECIFICATION
[SSS_DIL_GEN]

5.1 EULYNX SUBSYSTEM GENERIC IO [GEN-IO]

5.1.1 GENERAL REQUIREMENTS

The logical and functional system boundary of the EULYNX Subsystem Generic IO is shown in the following diagram.



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EL4-SSS-REQ-01494 Generic IO - Customer Specifications (Accepted)	
The EULYNX Subsystem Generic IO shall be implemented in accordance with [EULYNX Specification] - (EuDoc.45). The EULYNX Subsystem Generic IO shall implement all parameters which can be transmitted via SDI-IO.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.0 DCA_OBB_3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> <i>See also:</i>
<i>Verification:</i> Subit test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00518 Generic IO - SDI-IO - Operating Hours (Accepted)	
In addition to the data specified in the SDI-IO interface definition, the OC-IO shall count and transmit the operating hours in hours and decimal hours (h) of the OC via the same interface SDI-IO. This requirement refers to the total time under voltage of the Object Controller. The data shall be transmitted upon specific request and at an agreed frequency.	
<i>Comments:</i> Deviation: DCA can only transmit full hours and no decimal hours. Deviation is subject to customer approval.	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [09_19] <i>See also:</i>
<i>Verification:</i> Subit test	
<i>Issues/Decisions:</i>	

EL4-SSS-REQ-00998 Generic IO - OC Status Indication - Internal Elements (Accepted)	
The EULYNX Subsystem Generic IO shall support the following status indication with status LEDs: • Normal operation • Data traffic via ethernet interface • Maintenance/[fault] mode • Boot process • Initialisation mode	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [03_100] <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00524] Interfaces - IO5 - Maintainer
<i>Verification:</i> Subit test, System test	
<i>Issues/Decisions:</i>	

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EL4-SSS-REQ-01000		Generic IO - OC Status Indication - External Element		(Accepted)
The used ICB (ICB-IO or ICB-I2 or ICB-O2) for OC-IO shall indicate the status of the interface to the external elements locally with status LEDs: • Input ports: - i1) Input signal OK / no input signal - i2) (Fault) at the corresponding port • Output ports: - o1) Output switched on and OK / not switched on - o2) (Fault) at the corresponding port				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [03_100]	
Target Release:	DIL_OBB_1.0.x	See also:	==> [EL4-DIL-SSS-REQ-00524] Interfaces	
	DCA_OBB_3.1.0		- IO5 - Maintainer	
Verification:	Subit test, System test			
Issues/Decisions:				

5.1.2 INTERFACES

EL4-SSS-REQ-01432		Interfaces - SCI-IO - Command Interface		(Accepted)	
The EULYNX Subsystem Generic IO shall implement the interface SCI-IO according to the [EULYNX Specification].					
Comments:					
Category:		FUR		Priority:	
Originator:		Customer		Mandatory	
Target Release:		DIL_OBB_1.0.x		Origin:	
		DCA_OBB_3.1.0		==> [EL4 PH] [09_21]	
				See also:	
Verification:		Subit test			
Issues/Decisions:					

EL4-SSS-REQ-00520		Interfaces - SMI-IO - Maintainer		(Accepted)	
The EULYNX Subsystem Generic IO shall implement the interface SMI-IO according to the [EULYNX Specification].					
Comments:		Target Release for SMI is DCA 3.1.2. For DCA 3.1.0 Remote Download shall be delivered.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [09_22]		
Target Release:	DIL_OBB_2.0.0	See also:			
	DCA_OBB_3.1.2				
Verification:	Subit test				
Issues/Decisions:					

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EL4-SSS-REQ-00521		Interfaces - SDI-IO - Diagnosis		(Accepted)			
The EULYNX Subsystem Generic IO shall implement the interface SDI-IO according to the [EULYNX Specification].							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [09_18]	
Target Release:		DIL_OBB_1.0.x		See also:		==> [EL4 PH] [09_23]	
		DCA_OBB_3.1.0					
Verification:						Subit test	
Issues/Decisions:							

EL4-SSS-REQ-00522		Interfaces - SSI-IO - Security		(Accepted)			
The EULYNX Subsystem Generic IO shall implement the interface SSI-IO according to the [EULYNX Specification].							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [09_24]	
Target Release:		DIL_OBB_1.0.x		See also:			
		DCA_OBB_3.1.0					
Verification:		Subit test					
Issues/Decisions:							

EL4-SSS-REQ-00523		Interfaces - IO1 - Basic Data Identifier		(Accepted)			
The EULYNX Subsystem Generic IO shall implement the interface IO1 (Basic Data Identifier) according to the [EULYNX Specification].							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [09_25]	
Target Release:		DIL_OBB_1.0.x DCA_OBB_3.1.0		See also:			
Verification:		Subit test					
Issues/Decisions:							

EL4-SSS-REQ-00528		Interfaces - IO2 & 3 - Functional Logic		(Accepted)			
The [ECMUX] shall implement the functional logic for all elements using the interfaces IO2 and IO3, agreed with the customer dependent on the intended purposes.							
Comments:							
Category:		FUR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [09_31]	
Target Release:		DIL_OBB_1.0.x		See also:			
		DCA_OBB_3.1.0					
Verification:		Subit test					
Issues/Decisions:							

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6. SUBSYSTEM REQUIREMENTS FOR EULYNX SUBSYSTEM IO

6.1 EULYNX SUBSYSTEM GENERIC IO [GEN-IO]

6.1.1 GENERAL REQUIREMENTS

EL4-DFE-SSS-REQ-00438 IO - Serial Number (Draft)	
In addition to the data specified in the SDI-IO, the Object Controller-IO shall transmit the serial number of the Object Controller at the same interface in case of a successful registration in the customer network.	
<i>Comments:</i>	
<i>Category:</i> FUR	<i>Priority:</i> Mandatory
<i>Originator:</i> Technics (R&D)	<i>Origin:</i>
<i>Target Release:</i> DCA_OBB 3.1.0	<i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00462 Supported OC-IO-types - possible component connections (Draft)	
With one OC-IO (single-mode and also in master-slave-mode), different components (Line Keylock / Station Keylock / Local Control / Dispatch Lamp / Universal Element) are to be controlled or read back.	
<i>Comments:</i> OC-IO can be composed of different HW board (IC-IO, ICB-O2, IC-I2) depending on the required number of inputs and outputs to be managed.	
<i>Category:</i> FUR	<i>Priority:</i> Mandatory
<i>Originator:</i> Technics (R&D)	<i>Origin:</i>
<i>Target Release:</i> DFE_1.0.0 DCA_OBB 3.1.0	<i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

6.1.2 INTERFACES

6.1.2.1 [IO2] (GENERIC OUTPUT)

EL4-DFE-SSS-REQ-00392 Generic output interface (Draft)	
The [OC-IO] shall have a generic output interface IO2, which shall be configurable for universal purposes. A configuration as valent, antivalent or single output shall be possible for each output separately.	
<i>Comments:</i>	
<i>Category:</i> CSR	<i>Priority:</i> Mandatory
<i>Originator:</i> Customer	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00526] Interfaces - IO2 - Universal Outputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00528] Interfaces - IO2 & 3 - Functional Logic
<i>Target Release:</i> DFE_1.0.0 DCA_OBB 3.1.0	<i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

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EL4-DFE-SSS-REQ-00465 Generic output - connection loss (Draft)	
In case of a connection loss to the Interlocking Subsystem the [OC-IO] shall command all output ports to a safe state. The safe state (ON/OFF) shall be configurable via station data for each output port.	
<i>Comments:</i>	
<i>Category:</i> CSR <i>Originator:</i> Customer <i>Target Release:</i> DFE_1.0.0 DCA_OBB 3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00526] Interfaces - IO2 - Universal Outputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00528] Interfaces - IO2 & 3 - Functional Logic <i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00145 Generic output - interface specification (Draft)	
The outputs shall be able to switch voltages up to 400 VDC ($\pm 10\%$) or 400 VAC ($\pm 10\%$), 50 Hz and currents up to 3 A continuous current (12 A / 5s).	
<i>Comments:</i>	
<i>Category:</i> CSR <i>Originator:</i> Customer Technics (R&D) <i>Target Release:</i> DFE_1.0.0 DCA_OBB 3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00526] Interfaces - IO2 - Universal Outputs <i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

EL4-DFE-SSS-REQ-00466 OC-IO switching time output (Draft)	
The [OC-IO] shall have a maximum switching time of an output of 500ms (time from receiving the SCI command until the output changes).	
<i>Comments:</i> 500ms based on Safety Input - see also Safety Case ELEKTRA 2 - 3BU_15000_2327_DUAPC Ed08 - Appendix C.1.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> DFE_1.0.0 DCA_OBB 3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> <i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

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6.1.2.2 [IO3] (GENERIC INPUT)

EL4-DFE-SSS-REQ-00393		Generic input interface		(Draft)
The [OC-IO] shall have a generic input interface IO3, which shall be configurable for universal purposes. A configuration as valent (2-channel), antivalent (2-channel) or single input (1-channel) shall be possible for each input separately				
Comments:				
Category:	CSR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00528] Interfaces - IO2 & 3 - Functional Logic	
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0	See also:		
Verification:	Module test, Subit test			
Issues/Decisions:				

EL4-DFE-SSS-REQ-00464		Generic input - Day/Night sensor status read		(Draft)	
The [OC-IO] shall be able to read in information from a Day/Night sensor per station.					
Comments:		Open Point: Detail solution to be discussed. Current day/night sensor requires an analog input.			
Category:	CSR		Priority:	Mandatory	
Originator:	Customer		Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs	
Target Release:	DFE_1.0.0 DCA_OBB 3.1.0		See also:		
Verification:	Module test, Subit test				
Issues/Decisions:					

EL4-DFE-SSS-REQ-00146		Generic input - interface specification		(Draft)
The ICB inputs shall be configurable for voltages with nominal 24 VDC, 48 VDC and 60 VDC.				
Comments:		nominal voltages +/-20%		
Category:	CSR	Priority:	Mandatory	
Originator:	Customer Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs	
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0	See also:		
Verification:	Module test, Subit test			
Issues/Decisions:				

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EL4-DFE-SSS-REQ-00467 OC-IO detection time input (Draft)	
The [OC-IO] shall have a maximum detection time of an input of 500ms (time from changing the input until sending the SCI command).	
<i>Comments:</i> 500ms based on Safety Input - see also Safety Case ELEKTRA 2 - 3BU_15000_2327_DUAPC Ed08 - Appendix C.1.	
<i>Category:</i> FUR	<i>Priority:</i> Mandatory
<i>Originator:</i> Technics (R&D)	<i>Origin:</i>
<i>Target Release:</i> DCA_OBB_3.1.0	<i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

6.1.3 ICB-IO CONNECTION REQUIREMENTS

EL4-DFE-SSS-REQ-00190 Generic output - Potential-free safe switching contacts (relays) (Draft)	
Potential-free safe switching contacts (relays) or normally closed contacts (these are not supported by the DCA output) shall be implemented using an external relay box: The control of this relay box shall include the following: 1st path: Current measurement of the relay current 2nd path: Readback of a second relay switching contact with one input	
<i>Comments:</i>	
<i>Category:</i> CSR	<i>Priority:</i> Mandatory
<i>Originator:</i> Technics (R&D)	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00526] Interfaces - IO2 - Universal Outputs
<i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0	<i>See also:</i>
<i>Verification:</i> Module test, Subit test	
<i>Issues/Decisions:</i>	

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EL4-DFE-SSS-REQ-00182 Line Keylock (command-side) connection		(Draft)
A command-side Line Keylock shall be controlled with an OC-IO. Four secure outputs (2 x 4 output channels) are required for control and six input channels are required for readback and verify.		
ICB-IO:		
Secure Output 1:	60 V(+) - KLL_O1	
Secure Output 2:	60 V(-) - KLL_O1	
Secure Output 3:	60 V(+) - KLL_O2	
Secure Output 4:	60 V(-) - KLL_O2	
Input 1:	Readback KLL_O1	
Input 2:	Readback KLL_O2	
Input 3:	Readback KLL_I1	
Input 4:	Readback KLL_I2	
Input 5:	Readback KLL_I1	
Input 6:	Readback KLL_I2	
Input 7:	not connected	
Input 8:	not connected	
Comments:		
<i>Category:</i>	CSR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer Technics (R&D)	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements
<i>Target Release:</i>	DFE_1.0.0 DCA_OBB_3.1.0	<i>See also:</i>
<i>Verification:</i>	Module test, Subit test	
<i>Issues/Decisions:</i>		

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EL4-DFE-SSS-REQ-00460		Line Keylock (reporting side) connection		(Draft)
A reporting side Line Keylock shall be controlled with an OC-IO. Four inputs are required for keylock status read and verify.				
ICB-IO:				
Secure Output 1:	not connected			
Secure Output 2:	not connected			
Secure Output 3:	not connected			
Secure Output 4:	not connected			
Input 1:	not connected			
Input 2:	not connected			
Input 3:	Readback KLL_I1			
Input 4:	Readback KLL_I2			
Input 5:	Readback KLL_I1			
Input 6:	Readback KLL_I2			
Input 7:	not connected			
Input 8:	not connected			
ICB-I2:				
Input 1:	not connected			
Input 2:	not connected			
Input 3:	Readback KLL_I1			
Input 4:	Readback KLL_I2			
Input 5:	Readback KLL_I1			
Input 6:	Readback KLL_I2			
Input 7:	not connected			
Input 8:	not connected			
Input 9:	not connected			
Input 10:	not connected			
Input 11:	not connected			
Input 12:	not connected			
Input 13:	not connected			
Input 14:	not connected			
Input 15:	not connected			
Input 16:	not connected			
Comments:				
Category:	CSR	Priority:	Mandatory	
Originator:	Customer Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements	
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0	See also:		
Verification:	Module test, Subit test			
Issues/Decisions:				

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EL4-DFE-SSS-REQ-00183		Station Keylock connection		(Draft)			
A Station Keylock shall be controlled with an OC-IO. Four secure outputs (2 x 4 output channels) are required for control and four input channels are required for readback and verify.							
ICB-IO:							
Secure Output 1:		60 V(+) - KLS_O1					
Secure Output 2:		60 V(-) - KLS_O1					
Secure Output 3:		60 V(+) - KLS_O2					
Secure Output 4:		60 V(-) - KLS_O2					
Input 1:		Readback KLS_O1					
Input 2:		Readback KLS_O2					
Input 3:		Readback KLS_I1					
Input 4:		Readback KLS_I2					
Input 5:		not connected					
Input 6:		not connected					
Input 7:		not connected					
Input 8:		not connected					
Comments:							
Category:		CSR		Priority:		Mandatory	
Originator:		Customer Technics (R&D)		Origin:		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements	
Target Release:		DFE_1.0.0 DCA_OBB_3.1.0		See also:			
Verification:		Module test, Subit test					
Issues/Decisions:							

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EL4-DFE-SSS-REQ-00184 Local Operation connection		(Draft)
A Local Control for points and shunting signals shall be controlled with an OC-IO. Each Local Control requires one safe output (2 x 1 output channels) for control and two input channels for readback. Thus, up to four Local Control can be controlled with one OC-IO. ICB-IO: Secure Output 1: Local Op. 1_O Secure Output 2: Local Op. 2_O Secure Output 3: Local Op. 3_O Secure Output 4: Local Op. 4_O Input 1: Readback Local Op. 1_I1 Input 2: Readback Local Op. 1_I2 Input 3: Readback Local Op. 2_I1 Input 4: Readback Local Op. 2_I2 Input 5: Readback Local Op. 3_I1 Input 6: Readback Local Op. 3_I2 Input 7: Readback Local Op. 4_I1 Input 8: Readback Local Op. 4_I2		
Comments:		
<i>Category:</i> CSR <i>Originator:</i> Customer Technics (R&D) <i>Target Release:</i> DFE_1.0.0 DCA_OBB_3.1.0	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00884] 20250514 BG: tbd <i>See also:</i>	
<i>Verification:</i> Module test, Subit test		
<i>Issues/Decisions:</i>		

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EL4-DFE-SSS-REQ-00185		[Dispatch Lamp] connection 1		(Draft)
A [Dispatch Lamp] shall be controlled either with an OC-IO. Each [Dispatch Lamp] requires one safe output (2 x 1 output channels). Thus, up to 4 [Dispatch Lamp]s can be controlled with one ICB-IO and up to 8 [Dispatch Lamp]s with one ICB-O2.				
<u>ICB-IO:</u>				
Secure Output 1:	Dispatch Lamp 1			
Secure Output 2:	Dispatch Lamp 2			
Secure Output 3:	Dispatch Lamp 3			
Secure Output 4:	Dispatch Lamp 4			
Input 1:	not connected			
Input 2:	not connected			
Input 3:	not connected			
Input 4:	not connected			
Input 5:	not connected			
Input 6:	not connected			
Input 7:	not connected			
Input 8:	not connected			
<u>ICB-O2:</u>				
Secure Output 1:	Dispatch Lamp 1			
Secure Output 2:	Dispatch Lamp 2			
Secure Output 3:	Dispatch Lamp 3			
Secure Output 4:	Dispatch Lamp 4			
Secure Output 5:	Dispatch Lamp 5			
Secure Output 6:	Dispatch Lamp 6			
Secure Output 7:	Dispatch Lamp 7			
Secure Output 8:	Dispatch Lamp 8			
<i>Comments:</i>				
<i>Category:</i>	CSR	<i>Priority:</i>	Mandatory	
<i>Originator:</i>	Customer Technics (R&D)	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements	
<i>Target Release:</i>	DFE_1.0.0 DCA_OBB_3.1.0	<i>See also:</i>		
<i>Verification:</i>	Module test, Subit test			
<i>Issues/Decisions:</i>				

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EL4-DFE-SSS-REQ-00186		[Dispatch Lamp] connection 2		(Draft)
Free outputs of an ICB-IO (if fewer than 4 Local Control are controlled with one ICB-IO) shall be used to control Dispatch Lamps (2 x 1 output channel each).				
Comments:				
Category:	CSR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00527] Interfaces - IO3 - Universal Inputs ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01428] Usage - Element/Interface Logic - Elements	
Target Release:	DFE_1.0.0 DCA_OBB_3.1.0	See also:		
Verification:	Module test, Subit test			
Issues/Decisions:				

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7. HISTORY

7.1 EDITION 01

First version for DCA 3.1.0.

7.2 EDITION 02

Rework of Template.
Incorporated comments from DCA Review.
Added target version DCA 3.1.2. for multiple Requirements.
Multiple Requirements taken over from Top Level SSS to avoid Requirement inflation.
New requirement added: EL4-DFE-SSS-REQ-00474 - Transport Conditions

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

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